Pranav Joshi

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EDUCATION

Honours Bachelor of Mathematics/Business Administration (Co-op)

University of Waterloo - ON, Canada · Presidential Scholarship of Distinction · 2026 - 2031

SKILLS

- Technical: Python, JavaScript, SQL,
- Libraries & Frameworks: Flask, NumPy, PyTorch, Pandas
- Math: linear algebra, convex optimization, statistics and probability, Monte Carlo simulation, CVaR
- Cloud & AI: Google Cloud, Cohere API, OpenAI API, Gemini API, RAG, embeddings, vector search
- Familiar with: Gurobi, TensorFlow, React & React Native

EXPERIENCE

AI and Automation Contractor

Jaybird Studios — January 2026—February 2026, Vancouver, BC (remote)

- Engineered Python automation workflows integrating Gemini API, Google Cloud Functions, Zapier, and Make across customer registration and customer-data operations, saving an estimated 200+ hours/month and $9K annually.

AI and Automation Contractor

Lagree West — November 2025—January 2026, Vancouver, BC (remote)

- Optimized SQL queries and integrated the ChatGPT API into Google Cloud data pipelines for record processing and customer registration, generating an estimated $8K in annual savings.

Project Manager (Youth Innovation Program)

The Interac Corporation — September 2025—November 2025, Toronto, ON (hybrid)

- Conducted customer discovery to identify Gen Z financial pain points, synthesize user feedback, and establish user personas and product hypotheses around budgeting, saving, and financial decision-making.
- Translated research findings into product requirements, feature priorities, and an AI-powered financial assistant prototype; coordinated sprint execution with Interac mentors and presented product recommendations and the final business case to Interac executives.

AI & Automation Software Engineering Intern

Othership — May 2025—August 2025, Toronto, ON

- Engineered and deployed a production account-merging workflow integrating the MarianaTek API, consolidating duplicate customer accounts across a system handling 150K+ accounts annually and reducing manual reconciliation.
- Built and deployed GPT/RAG applications and Google Cloud automation pipelines, including an AI support bot and a 3,000-page lease-analysis system, contributing to $12K+ in annual savings and 300+ hours saved per year.

PROJECTS

Budgety: AI Powered Financial Assistant | *Python, Flask, JavaScript, Cohere API*

- Architected a Flask backend integrating Cohere LLM inference with an embedding-based vector search pipeline, retrieving relevant financial context from user profiles, transaction histories, budgets, and savings goals to ground personalized responses.
- Designed the MVP in a 3-week sprint and presented the final prototype to Interac executives at their Toronto headquarters.

Inventory Optimization (Experiment and Simulation) | *Python, Operations Research Mathematics*

- Programmed a classic production and inventory optimization problem (where setup and holding costs must be balanced against randomized demand over time) and tested the advantages of an algorithm based on the Wagner-Whitin model over the baseline "Lot-for-Lot" method (producing supply only when obviously needed).
- Achieved an average ~22.8% cost reduction using the optimized Wagner-Whitin algorithm versus the baseline strategy, with optimality validated against brute force across 999 randomized scenarios (<700 ms total runtime).

Reinsurance Pricing Optimization | *Python, Actuarial Science, Risk Analysis, Convex Optimization*

- Simulated catastrophe loss events for regional insurers and their reinsurance claims via Monte Carlo simulation.
- Minimized expected reinsurance cost subject to a 95% CVaR constraint (bounding the expected loss in the worst 5% of scenarios), formulating the tail-risk objective as a linear program via the Rockafellar–Uryasev reformulation.
- Undercut a baseline pricing model by ~1.5% in expected cost using the Rockafellar–Uryasev reformulation, translating to a ~8.5% reduction in customer premiums while holding tail risk within the 95% CVaR constraint.

ML-Driven Autonomous Vision-Guided Robot | *Python, YOLOv8-seg, OpenCV, PyTorch, Arduino*

- Trained a custom YOLOv8-seg instance-segmentation model (on an RTX 4050) to segment the driving path from a live video feed in real time, from ~6 minutes of demonstration footage.
- Ran inference on a host laptop that converted segmentation output into motor-control commands streamed to an Arduino, closing the perception-to-actuation loop; the robot autonomously completed the course.
- Achieved 344/360 correct path segments (95.6%) on a 30-second test feed at 12 fps.